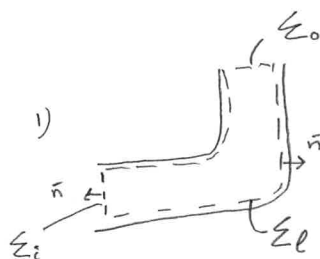
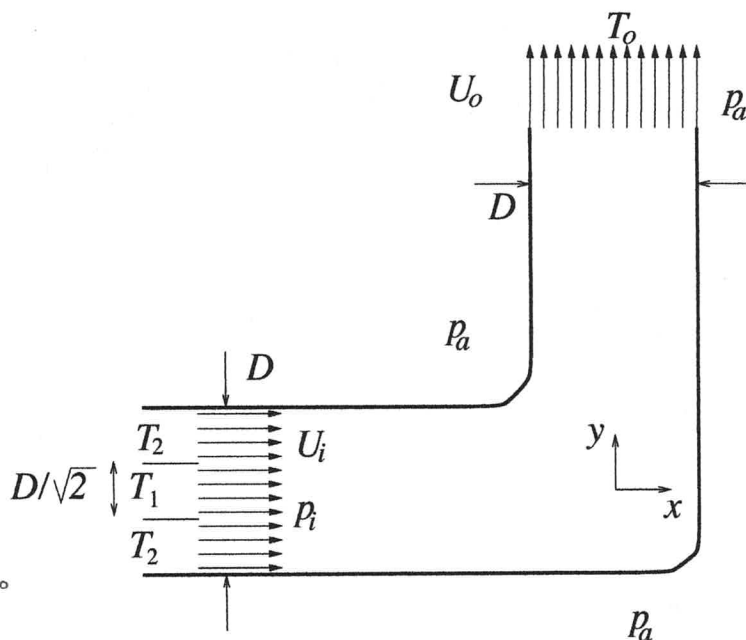


The elbow-shaped tube of circular cross section shown in the figure serves to mix two different air streams. The pipe discharges to the open atmosphere after mixing is completed. At the inlet, where the pressure $p_i = 2p_a$ and velocity U_i are uniform, the inner stream, with diameter $D/\sqrt{2}$, has temperature T_1 , and the outer coflow stream has temperature T_2 . The temperature measurement at the outlet reveals that $T_o = T_1 T_2 / (T_1 + T_2)$.

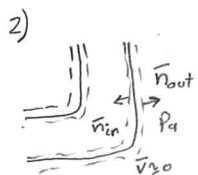
1. Determine the uniform gas velocity at the outlet U_o .
2. Compute the force $\bar{F} = F_x \bar{e}_x + F_y \bar{e}_y$ acting on the pipe. In the computation, neglect the effect of gravity.
3. Obtain the amount of heat transferred per unit time through the pipe wall.

Note: For a perfect gas, $p = \rho R_g T$, where R_g is the gas constant.



$$\int_{\Sigma_c} \bar{v} \cdot \bar{n} d\sigma = 0 = -\frac{P_i}{R_g T_1} U_i \frac{\pi D^2}{8} - \frac{P_i}{R_g T_2} U_i \frac{\pi}{4} (1 - \frac{1}{2}) D^2 + \frac{P_a}{R_g T_o} U_o \frac{\pi}{4} D^2 \Rightarrow \boxed{U_o = U_i}$$

$P_i = 2P_a$
 $\frac{1}{T_o} = \frac{1}{T_1} + \frac{1}{T_2}$



$$\bar{F} = -\int_{\Sigma_{in} + \Sigma_{out}} p \bar{n} d\sigma + \int_{\Sigma_{in} + \Sigma_{out}} \bar{E} \cdot \bar{n} d\sigma = -\int_{\Sigma_{in}} (p - P_a) \bar{n}_{in} d\sigma + \int_{\Sigma_{in}} \bar{E} \cdot \bar{n}_{in} d\sigma$$

USING THE CONTROL VOLUME SHOWN IN 1)

$$\begin{aligned} \int_{\Sigma_c} \bar{v} \cdot \bar{n} d\sigma &= -\int_{\Sigma_c} (p - P_a) \bar{n} d\sigma + \int_{\Sigma_c} \bar{E} \cdot \bar{n} d\sigma \\ &= -\frac{P_i}{R_g T_1} U_i U_i \bar{e}_x \frac{\pi D^2}{8} - \frac{P_i}{R_g T_2} U_i U_i \bar{e}_x \frac{\pi D^2}{8} + \frac{P_a}{R_g T_o} U_o U_o \bar{e}_y \frac{\pi D^2}{4} \bar{e}_y = (P_i - P_a) \frac{\pi D^2}{4} \bar{e}_x - \underbrace{\int_{\Sigma_l} (p - P_a) \bar{n} d\sigma + \int_{\Sigma_l} \bar{E} \cdot \bar{n} d\sigma}_{-\bar{F}} \end{aligned}$$

$$\bar{F} = \frac{\pi D^2}{4} P_a \left(1 + \frac{U_i^2}{R_g T_o} \right) \bar{e}_x + \frac{\pi D^2}{4} P_a \frac{U_i^2}{R_g T_o} \bar{e}_y$$

3)

$$\int_{\Sigma_c} \rho \left(\bar{e} + \frac{v^2}{2} \right) \bar{v} \cdot \bar{n} d\sigma = -\int_{\Sigma_c} p \bar{v} \cdot \bar{n} d\sigma + \int_{\Sigma_c} \bar{v} \cdot \bar{E} \cdot \bar{n} d\sigma - \int_{\Sigma_c} \bar{q} \cdot \bar{n} d\sigma \Rightarrow \dot{Q}_{\text{FROM WALL TO FLUID}} = -\int_{\Sigma_l} \bar{q} \cdot \bar{n} d\sigma = \int_{\Sigma_l} \rho \left(\bar{h} + \frac{v^2}{2} \right) \bar{v} \cdot \bar{n} d\sigma = -\frac{P_a U_i A C_p}{R_g}$$

↓
THE LIQUID IS COOLED DOWN
AS IT FLOWS THROUGH THE PIPE